

Table 2 – Class 2 stranded conductors for single-core and multi-core cables

1	2	3	4	5	6	7	8	9	10
Nominal cross-sectional area mm ²	Minimum number of wires in the conductor						Maximum resistance of conductor at 20°C		
	Circular		Circular compacted		Shaped		Annealed copper conductor		Aluminium or aluminium alloy conductor ^c
	Cu	Al	Cu	Al	Cu	Al	Plain wires	Metal-coated wires	
							Ω/km	Ω/km	Ω/km
0,5	7	-	-	-	-	-	36,0	36,7	-
0,75	7	-	-	-	-	-	24,5	24,8	-
1,0	7	-	-	-	-	-	18,1	18,2	-
1,5	7	-	6	-	-	-	12,1	12,2	-
2,5	7	-	6	-	-	-	7,41	7,56	-
4	7	-	6	-	-	-	4,61	4,70	-
6	7	-	6	-	-	-	3,08	3,11	-
10	7	7	6	6	-	-	1,83	1,84	3,08
16	7	7	6	6	-	-	1,15	1,16	1,91
25	7	7	6	6	6	6	0,727	0,734	1,20
35	7	7	6	6	6	6	0,524	0,529	0,868
50	19	19	6	6	6	6	0,387	0,391	0,641
70	19	19	12	12	12	12	0,268	0,270	0,443
95	19	19	15	15	15	15	0,193	0,195	0,320
120	37	37	18	15	18	15	0,153	0,154	0,253
150	37	37	18	15	18	15	0,124	0,126	0,206
185	37	37	30	30	30	30	0,0991	0,100	0,164
240	37	37	34	30	34	30	0,0754	0,0762	0,125
300	61	61	34	30	34	30	0,0601	0,0607	0,100
400	61	61	53	53	53	53	0,0470	0,0475	0,0778
500	61	61	53	53	53	53	0,0366	0,0369	0,0605
630	91	91	53	53	53	53	0,0283	0,0286	0,0469
800	91	91	53	53	-	-	0,0221	0,0224	0,0367
1 000	91	91	53	53	-	-	0,0176	0,0177	0,0291
1 200	b						0,0151	0,0151	0,0247
1 400 ^a	b						0,0129	0,0129	0,0212
1 600	b						0,0113	0,0113	0,0186
1 800 ^a	b						0,0101	0,0101	0,0165
2 000	b						0,0090	0,0090	0,0149
2 500	b						0,0072	0,0072	0,0127

^a These sizes are non-preferred. Other non-preferred sizes are recognized for some specialized applications but are not within the scope of this standard.

^b The minimum number of wires for these sizes is not specified. These sizes may be constructed from 4, 5 or 6 equal segments (Milliken).

^c For stranded aluminium alloy conductors having the same nominal cross-sectional area as an aluminium conductor the resistance value should be agreed between the manufacturer and the purchaser.

Table 3 – Class 5 flexible copper conductors for single core and multi-core cables

1	2	3	4
Nominal cross-sectional area mm ²	Maximum diameter of wires in conductor mm	Maximum resistance of conductor at 20 °C	
		Plain wires Ω/km	Metal-coated wires Ω/km
0,5	0,21	39,0	40,1
0,75	0,21	26,0	26,7
1,0	0,21	19,5	20,0
1,5	0,26	13,3	13,7
2,5	0,26	7,98	8,21
4	0,31	4,95	5,09
6	0,31	3,30	3,39
10	0,41	1,91	1,95
16	0,41	1,21	1,24
25	0,41	0,780	0,795
35	0,41	0,554	0,565
50	0,41	0,386	0,393
70	0,51	0,272	0,277
95	0,51	0,206	0,210
120	0,51	0,161	0,164
150	0,51	0,129	0,132
185	0,51	0,106	0,108
240	0,51	0,0801	0,0817
300	0,51	0,0641	0,0654
400	0,51	0,0486	0,0495
500	0,61	0,0384	0,0391
630	0,61	0,0287	0,0292

Annex A **(normative)**

Measurement of resistance

The cable shall be kept in the test area for sufficient time to ensure that the conductor temperature has reached a level which permits an accurate determination of resistance using the correction factors provided.

Measure the d.c. resistance of the conductor(s), either on a complete length of cable or flexible cord or on a sample of cable or flexible cord of at least 1 m in length, at room temperature and record the temperature at which the measurement is made. Adjust the measured resistance by means of the correction factors given in Table A.1.

Calculate the resistance per kilometre length of cable from the length of the complete cable and not from the length of the individual core or wires.

If necessary, correction to 20 °C and 1 km length shall be made by applying the following formula:

$$R_{20} = R_t \times k_t \times \frac{1\,000}{L}$$

where

k_t is the temperature correction factor from Table A.1;

R_{20} is the conductor resistance at 20 °C, in Ω/km ;

R_t is the measured conductor resistance, in Ω ;

L is the length of the cable, in m.

Table A.1 – Temperature correction factors k_t for conductor resistance to correct the measured resistance at t °C to 20 °C

1	2	1	2
Temperature of conductor at time of measurement t °C	Correction factor, k_t All conductors	Temperature of conductor at time of measurement t °C	Correction factor, k_t All conductors
0	1,087	21	0,996
1	1,082	22	0,992
2	1,078	23	0,988
3	1,073	24	0,984
4	1,068	25	0,980
5	1,064	26	0,977
6	1,059	27	0,973
7	1,055	28	0,969
8	1,050	29	0,965
9	1,046	30	0,962
10	1,042	31	0,958
11	1,037	32	0,954
12	1,033	33	0,951
13	1,029	34	0,947
14	1,025	35	0,943
15	1,020	36	0,940
16	1,016	37	0,936
17	1,012	38	0,933
18	1,008	39	0,929
19	1,004	40	0,926
20	1,000		

NOTE The values of correction factors k_t are based on a resistance-temperature coefficient of 0,004 per K at 20 °C.

The values of temperature correction factors specified in column 2 are approximate but give practical values well within the accuracy that can normally be achieved in the measurements of conductor temperature and length of cables or flexible cords.

For more accurate values for the temperature correction factors for copper and aluminium, reference should be made to Annex B. However, these should not be treated as a requirement for testing in compliance with this standard in the assessment of resistances.

Annex B

(informative)

Exact formulae for the temperature correction factors

a) Annealed copper conductors: plain or metal coated

$$k_{t,Cu} = \frac{254,5}{234,5 + t} = \frac{1}{1 + 0,00393 (t - 20)}$$

b) Aluminium conductors

$$k_{t,Al} = \frac{248}{228 + t} = \frac{1}{1 + 0,00403 (t - 20)}$$

NOTE For aluminium alloys, reference should be made to the manufacturer.

In all the above cases, t refers to the temperature of the conductor at the time of measurement in degrees Celsius.

**Table C.1 – Maximum diameters of circular copper conductors –
solid, non-compacted stranded and flexible**

1	2	3	4
Cross sectional area mm ²	Conductors in cables for fixed installations		Flexible conductors (Classes 5 and 6) mm
	Solid (Class 1) mm	Stranded (Class 2) mm	
0,5	0,9	1,1	1,1
0,75	1,0	1,2	1,3
1,0	1,2	1,4	1,5
1,5	1,5	1,7	1,8
2,5	1,9	2,2	2,4
4	2,4	2,7	3,0
6	2,9	3,3	3,9
10	3,7	4,2	5,1
16	4,6	5,3	6,3
25 ^a	5,7	6,6	7,8
35 ^a	6,7	7,9	9,2
50 ^a	7,8	9,1	11,0
70 ^a	9,4	11,0	13,1
95 ^a	11,0	12,9	15,1
120 ^a	12,4	14,5	17,0
150 ^a	13,8	16,2	19,0
185	15,4	18,0	21,0
240	17,6	20,6	24,0
300	19,8	23,1	27,0
400	22,2	26,1	31,0
500	-	29,2	35,0
630	-	33,2	39,0
800	-	37,6	-
1 000	-	42,2	-
NOTE The values given for flexible conductors are intended to allow for both class 5 and class 6 conductors.			
^a See 5.1.1 b).			

Table C.2 – Minimum and maximum diameters of stranded compacted circular copper, aluminium and aluminium alloy conductors

1	2	3
Cross-sectional area mm ²	Stranded compacted circular conductors (Class 2)	
	Minimum diameter mm	Maximum diameter mm
10	3,6	4,0
16	4,6	5,2
25	5,6	6,5
35	6,6	7,5
50	7,7	8,6
70	9,3	10,2
95	11,0	12,0
120	12,3	13,5
150	13,7	15,0
185	15,3	16,8
240	17,6	19,2
300	19,7	21,6
400	22,3	24,6
500	25,3	27,6
630	28,7	32,5
NOTE 1 The dimensional limits of aluminium conductors with cross-sectional areas above 630 mm ² are not given as the compaction technology is not generally established.		
NOTE 2 No values are given for compacted copper conductors in the size range 1,5 mm ² to 6 mm ² .		